Quantitative Comparisons

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

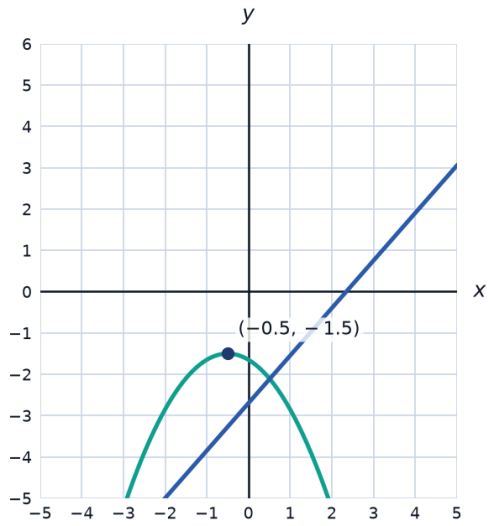
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Numeric comparisons

Column A: 7×8. Column B: $6 \times 9 + 2$. Compare the two columns: which is greater, are they equal, or is it impossible to determine?

2 [Foundational] • Target: Numeric comparisons

Column A: $123 + 456$. Column B: $700 - 120$. Compare the two columns.



3 [Foundational] • Target: Percent comparisons

Column A: 25% of 160. Column B: 20% of 200. Compare the two columns.

4 [Foundational] • Target: Percent comparisons

Column A: the result of increasing 100 by 10% twice in a row. Column B: the result of increasing 100 by 20% once. Compare the two columns.

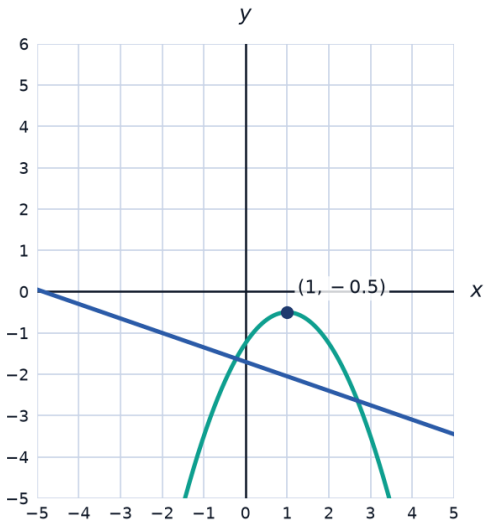
5 [Foundational] • Target: Algebraic comparisons at a fixed value

Given $x = 5$. Column A: $2x + 3$. Column B: $x + 8$. Compare the two columns.

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6 [Foundational] • Target: Comparisons requiring testing values

For any real number x: Column A: x. Column B: x^2. Compare the two columns.



7 [Foundational] • Target: Comparisons requiring testing values

Given only that n is an integer: Column A: n^2. Column B: 4. Compare the two columns.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Geometry comparisons

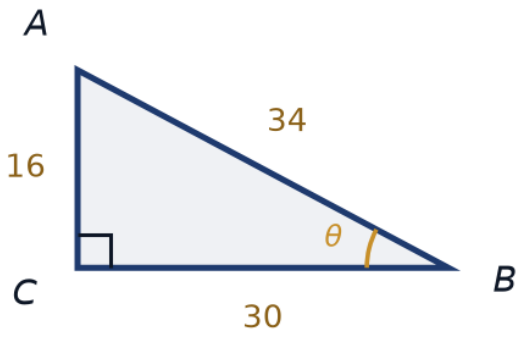
Column A: the perimeter of a square with side 6 cm. Column B: the perimeter of a rectangle with length 8 cm and width 3 cm. Compare the two columns.

9 [Intermediate] • Target: Geometry comparisons

Column A: the area of a circle with radius 4 cm (use $\pi \approx 3.14$). Column B: the area of a square with side 7 cm. Compare the two columns.

10 [Intermediate] • Target: Geometry comparisons

Column A: the sum of the interior angles of a pentagon. Column B: the sum of the interior angles of two triangles. Compare the two columns.



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11 [Intermediate] • Target: Geometry comparisons

Column A: the hypotenuse of a right triangle with legs 5 and 12. Column B: 14. Compare the two columns.

12 [Intermediate] • Target: Ratio and proportion comparisons

Column A: the value of x, if $x : 12 = 3 : 4$. Column B: 10. Compare the two columns.

13 [Intermediate] • Target: Ratio and proportion comparisons

A recipe uses flour and sugar in the ratio $3 : 1$. Column A: the sugar needed for 240 g of flour. Column B: 90 g. Compare the two columns.

14 [Intermediate] • Target: Ratio and proportion comparisons

Column A: the sum of the parts in the simplified form of the ratio $18 : 24$. Column B: 8. Compare the two columns.

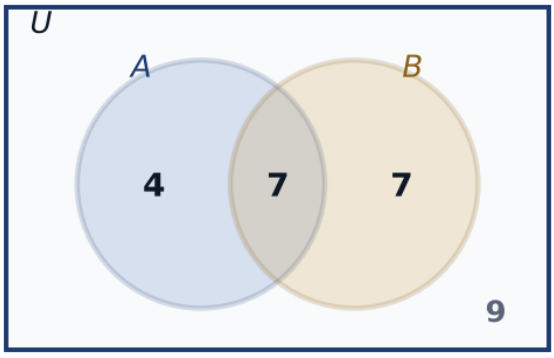
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Ratio and proportion comparisons • Cross-Lesson: 3.01 Algebra Essentials

Two numbers are in the ratio $5 : 3$, and their sum is 96. Column A: the larger number. Column B: 61. Compare the two columns.

16 [Challenging] • Target: Data and probability comparisons • Cross-Lesson: 4.01 Geometry & Measurement

A data set is $4, 7, 7, 9, 13$. Column A: the mean. Column B: the median. Compare the two columns.



17 [Challenging] • Target: Data and probability comparisons • Cross-Lesson: 6.01 Sequences & Patterns

Column A: the range of the data set $3, 8, 15, 22$. Column B: 20. Compare the two columns.

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18 [Challenging] • Target: Data and probability comparisons • Cross-Lesson: 7.01 Data Analysis & Probability

A bag contains 3 red balls and 5 blue balls. Column A: the probability of drawing a red ball. Column B: the probability of drawing a blue ball. Compare the two columns.

19 [Challenging] • Target: Word-problem comparisons • Cross-Lesson: 8.01 Word Problems & Reasoning

Column A: the total cost of 4 items at 15 riyals each, after a 10% discount on the total. Column B: 55 riyals. Compare the two columns.

20 [Challenging] • Target: Word-problem comparisons • Cross-Lesson: 1.01 Arithmetic & Number Sense

Ali's age is 3 years more than twice Sara's age. Sara is 8. Column A: Ali's age. Column B: 20. Compare the two columns.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Percent comparisons

Column A: the result of decreasing 100 by 20%, then increasing that result by 20%. Column B: 96.

a. The relationship cannot be determined from the given information

b. The two columns are equal

c. Column A is greater

d. Column B is greater

22 [Foundational] • Target: MCQ — Comparisons requiring testing values

For any integer n: Column A: n. Column B: $-n$. Compare the two columns.

a. The two columns are equal

b. The relationship cannot be determined from the given information

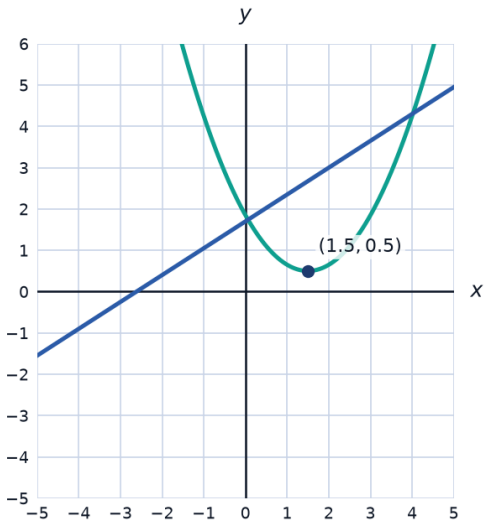
c. Column B is greater

d. Column A is greater

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23 [Foundational] • Target: MCQ — Comparisons requiring testing values

Given only that m is a positive integer: Column A: m. Column B: m^2. Compare the two columns.



a. Column B is greater

b. Column A is greater

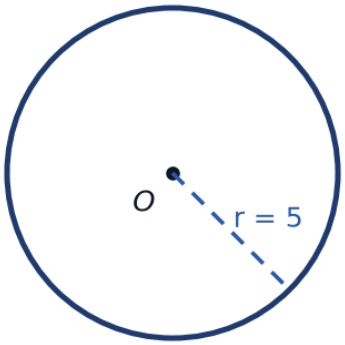
c. The two columns are equal

d. The relationship cannot be determined from the given information

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24 [Foundational] • Target: MCQ — Geometry comparisons

Column A: the circumference of the circle shown (use $\pi \approx 3.14$). Column B: 32.



a. The relationship cannot be determined from the given information

b. Column A is greater

c. The two columns are equal

d. Column B is greater

25 [Foundational] • Target: MCQ — Geometry comparisons

Column A: the area of the rectangle shown. Column B: the perimeter of a square with side 7.

a. The relationship cannot be determined from the given information

b. The two columns are equal

c. Column B is greater

d. Column A is greater

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26 [Foundational] • Target: MCQ — Geometry comparisons

Column A: the sum of the interior angles of a hexagon. Column B: the sum of the interior angles of two triangles.

a. The two columns are equal

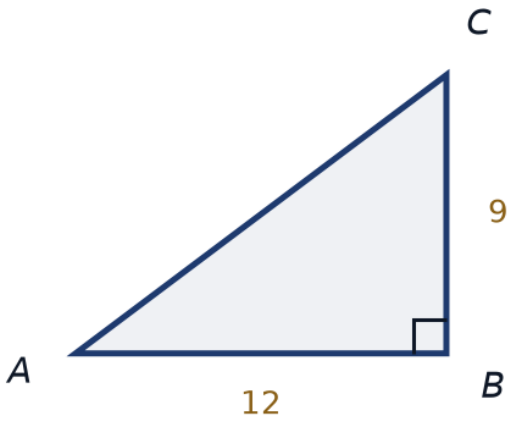
b. The relationship cannot be determined from the given information

c. Column A is greater

d. Column B is greater

27 [Foundational] • Target: MCQ — Geometry comparisons

Column A: the hypotenuse of the right triangle shown. Column B: 16.



a. The relationship cannot be determined from the given information

b. Column B is greater

c. The two columns are equal

d. Column A is greater

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28 [Foundational] • Target: MCQ — Geometry comparisons

Column A: the volume of a cylinder with radius 3 and height 7 (use $\pi \approx 3.14$). Column B: 200.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. The two columns are equal

d. Column B is greater

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Ratio and proportion comparisons

Column A: the value of x, if $x : 15 = 2 : 5$. Column B: 7.

a. The relationship cannot be determined from the given information

b. The two columns are equal

c. Column A is greater

d. Column B is greater

30 [Intermediate] • Target: MCQ — Ratio and proportion comparisons

Two numbers are in the ratio $4 : 7$, and their sum is 99. Column A: the smaller number. Column B: 40.

a. The two columns are equal

b. The relationship cannot be determined from the given information

c. Column A is greater

d. Column B is greater

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31 **[Intermediate]** • **Target: MCQ — Ratio and proportion comparisons**

Column A: the sum of the parts in the simplified form of the ratio $24 : 36$. Column B: 5.

a. Column B is greater

b. The relationship cannot be determined from the given information

c. Column A is greater

d. The two columns are equal

32 **[Intermediate]** • **Target: MCQ — Ratio and proportion comparisons**

A recipe uses water and rice in the ratio $2 : 1$. Column A: the water needed for 300 g of rice. Column B: 550 g.

a. Column B is greater

b. Column A is greater

c. The two columns are equal

d. The relationship cannot be determined from the given information

33 **[Intermediate]** • **Target: MCQ — Data and probability comparisons**

A data set is $5, 8, 8, 10, 14$. Column A: the mean. Column B: the median.

a. The two columns are equal

b. Column B is greater

c. Column A is greater

d. The relationship cannot be determined from the given information

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34 **[Intermediate]** • **Target: MCQ — Data and probability comparisons**

Column A: the range of the data set $6, 11, 19, 25$. Column B: 19.

a. The relationship cannot be determined from the given information

b. Column B is greater

c. The two columns are equal

d. Column A is greater

35 **[Intermediate]** • **Target: MCQ — Data and probability comparisons**

A bag contains 4 green balls and 9 yellow balls. Column A: the probability of drawing a green ball. Column B: 0.4.

a. Column B is greater

b. The relationship cannot be determined from the given information

c. The two columns are equal

d. Column A is greater

36 **[Intermediate]** • **Target: MCQ — Data and probability comparisons**

A data set is $2, 4, 4, 4, 6$. Column A: the mode. Column B: the mean.

a. Column A is greater

b. The relationship cannot be determined from the given information

c. The two columns are equal

d. Column B is greater

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37 [Intermediate] • Target: MCQ — Word-problem comparisons

Column A: the total cost of 5 items at 20 riyals each, after a 15% discount on the total. Column B: 90.

a. The two columns are equal

b. Column A is greater

c. The relationship cannot be determined from the given information

d. Column B is greater

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Word-problem comparisons • Cross-Lesson: 7.01 Data Analysis & Probability

Layla's age is 4 years less than three times Huda's age. Huda is 9. Column A: Layla's age. Column B: 25.

a. Column A is greater

b. The two columns are equal

c. Column B is greater

d. The relationship cannot be determined from the given information

39 [Challenging] • Target: MCQ — Word-problem comparisons • Cross-Lesson: 8.01 Word Problems & Reasoning

Column A: the simple interest on 3,000 riyals at 4% per year for 2 years. Column B: 250.

a. The two columns are equal

b. The relationship cannot be determined from the given information

c. Column A is greater

d. Column B is greater

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40 [Challenging] • Target: MCQ — Word-problem comparisons • Cross-Lesson: 1.01 Arithmetic & Number Sense

A car travels 300 km using 25 liters of fuel. Column A: the distance travelled per liter. Column B: 12 km.

a. The two columns are equal

b. The relationship cannot be determined from the given information

c. Column B is greater

d. Column A is greater

41 [Challenging] • Target: MCQ — Comparisons with inequalities and ranges • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Given that $3 < x < 8$. Column A: x. Column B: 5.

a. The relationship cannot be determined from the given information

b. The two columns are equal

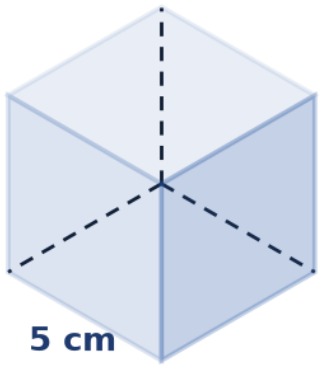
c. Column A is greater

d. Column B is greater

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42 [Challenging] • Target: MCQ — More geometry comparisons • Cross-Lesson: 3.01 Algebra Essentials

Column A: the volume of the cube shown. Column B: the volume of a rectangular box $4 \times 5 \times 7$.



a. The relationship cannot be determined from the given information

b. Column A is greater

c. Column B is greater

d. The two columns are equal

43 [Challenging] • Target: MCQ — More geometry comparisons • Cross-Lesson: 4.01 Geometry & Measurement

Column A: the circumference of a circle with radius 7 cm (use $\pi = \frac{22}{7}$). Column B: the area of a square with side 11 cm.

a. The two columns are equal

b. Column A is greater

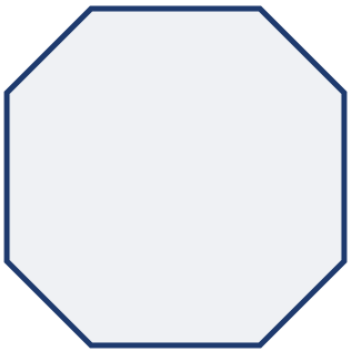
c. Column B is greater

d. The relationship cannot be determined from the given information

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44 [Challenging] • Target: MCQ — More geometry comparisons • Cross-Lesson: 6.01 Sequences & Patterns

Column A: the sum of the interior angles of the octagon shown. Column B: four times the sum of the interior angles of a triangle.



a. Column B is greater

b. The two columns are equal

c. Column A is greater

d. The relationship cannot be determined from the given information

45 [Challenging] • Target: Numeric comparisons • Cross-Lesson: 7.01 Data Analysis & Probability

For any real x. Column A: x^2. Column B: 4. Compare the two columns.

a. Column B is greater

b. Column A is greater

c. The relationship cannot be determined from the given information

d. The two columns are equal

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